

CHAPTER 1

INTRODUCTION

1.1 MACHINE LEARNING IN HEALTHCARE

Machine learning has transformed healthcare by enabling computers to identify patterns across thousands of patient records simultaneously, complementing clinical decision-making in ways that address the limitations of human capacity and time. Cardiovascular disease remains one of the leading causes of mortality worldwide, and its risk assessment involves multiple interacting variables including blood pressure, cholesterol, body weight, age, and lifestyle factors, making it well suited for predictive modelling. Early and accurate detection is critical, as timely intervention significantly improves patient outcomes and reduces the burden on healthcare systems. However, machine learning models in healthcare carry unique responsibilities. A model that performs well statistically but behaves irresponsibly in uncertain situations can cause real harm, driving the need for systems that are not only accurate but honest about their limitations, flagging borderline cases for human review rather than producing unreliable automated decisions.

1.2 OBJECTIVES

The first objective is to develop a robust machine learning pipeline capable of predicting cardiovascular disease from patient health records with high accuracy and clinical relevance, using the Kaggle Cardiovascular Disease dataset of approximately 70,000 records. The second objective is to apply comprehensive feature engineering techniques including clinical domain metrics, polynomial interactions, target encoding, dimensionality reduction through PCA, and patient archetype identification through K-Means clustering, in order to extract maximum predictive signal from the available data. The third objective is to construct a high-performance stacking ensemble combining XGBoost, CatBoost, and LightGBM as base learners with a Logistic Regression meta-learner, trained through stratified cross-validation to produce reliable and unbiased probability estimates. The fourth objective is to design and implement a Clinical Confidence Rejection mechanism that differentiates between high-confidence and borderline predictions, routing uncertain cases to medical professionals rather than forcing unreliable autonomous decisions. The fifth objective is to demonstrate that a medically responsible AI system, one that acknowledges its own uncertainty, can achieve superior practical performance compared to conventional classification models that optimize purely for statistical metrics without regard for real-world clinical deployment.

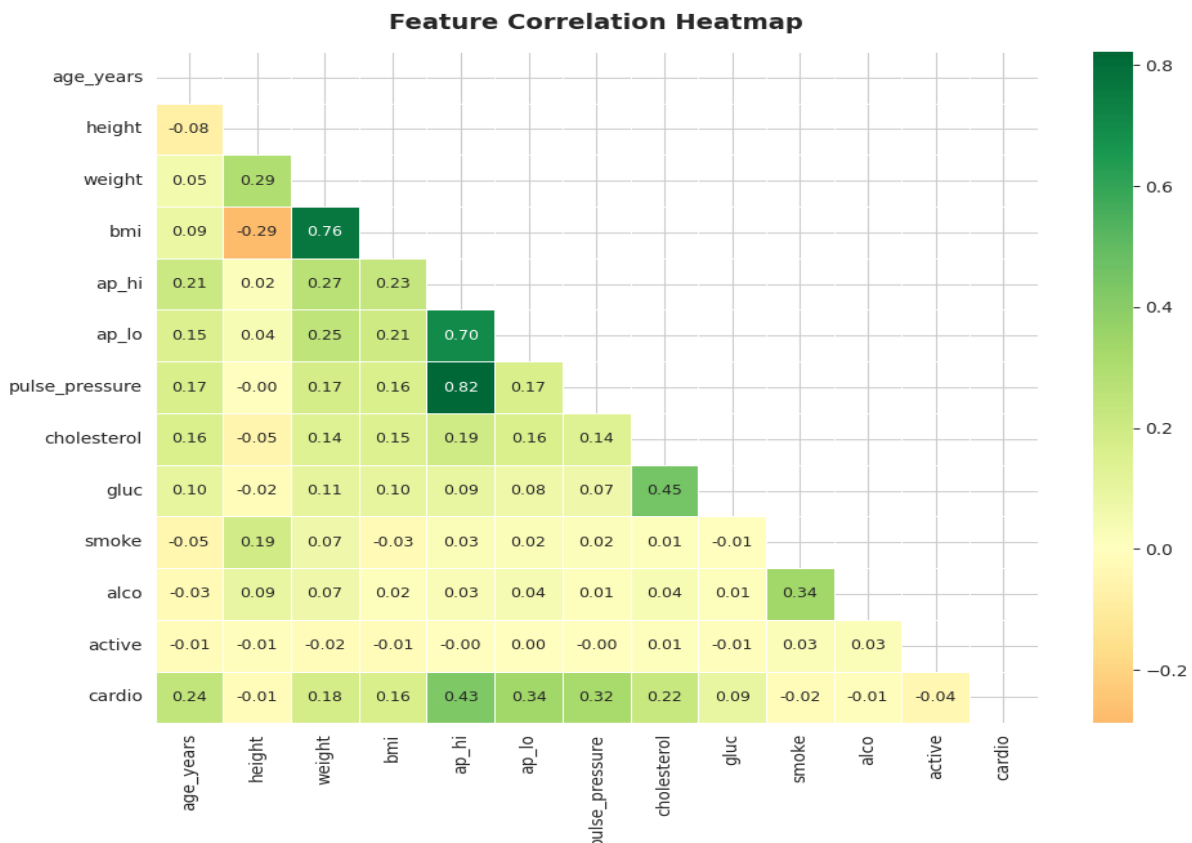
CHAPTER 2

DATASET

2.1 CARDIOVASCULAR DISEASE DATASET

Cardiovascular Disease dataset contains 70,000 patient records collected from routine medical examinations, each described by 13 attributes across three categories. Objective features include age in days, height, weight, and gender. Examination features include systolic blood pressure, diastolic blood pressure, cholesterol, and glucose, where cholesterol and glucose are ordinal values on a three-point scale representing normal, above normal, and well above normal. Subjective features are self-reported and cover smoking status, alcohol consumption, and physical activity as binary indicators. The target variable is binary, with 1 indicating cardiovascular disease and 0 indicating its absence.

The class distribution is nearly balanced at 49.97 percent diseased and 50.03 percent healthy, making standard accuracy metrics reliable without aggressive resampling. Patient ages range from 29 to 65 years with the majority between 45 and 60, consistent with clinical cardiovascular risk patterns. Entries with physiologically impossible blood pressure values were removed during preprocessing. Despite its size, the dataset carries an inherent accuracy ceiling of approximately 73.5 percent for conventional classifiers, as the 11 features provide insufficient discriminative power for a portion of genuinely ambiguous cases, which directly motivated the Clinical Confidence Rejection mechanism central to this project.



Some Sample Data Representaions:



CHAPTER 3

DEEP LEARNING

MACHINE LEARNING MODEL

To establish a scientifically robust benchmark for this study, initial experimentation evaluated "pure" machine learning workflows using standard classifiers. We utilized a physiology-enriched dataset featuring engineered variables like Body Mass Index (BMI), Pulse Pressure, and Mean Arterial Pressure (MAP) to provide the models with direct clinical context. To ensure the evaluation was statistically sound and free of data leakage, these baseline models were tested via rigorous 5-Fold Stratified Cross-Validation without the safety net of a clinical rejection option.

We first tested **Logistic Regression**, which yielded an average Accuracy of 0.7288 and a Recall of 0.6737. While highly interpretable, its linear decision boundary failed to capture the complex, overlapping biological markers of cardiovascular disease. Next, we deployed **XGBoost** to map non-linear feature interactions. Although XGBoost improved performance—reaching an Accuracy of 0.7364 and an ROC-AUC of 0.8027—it remained fundamentally constrained by what we identified as the dataset's mathematical ceiling of approximately 73.5%.

Baseline 5-Fold Cross-Validation Results:

- **Logistic Regression:** Accuracy: 0.7288±0.0031 | ROC-AUC: 0.7926±0.0031 | Recall: 0.6737±0.0059
- **XGBoost:** Accuracy: 0.7364±0.0022 | ROC-AUC: 0.8027±0.0030 | Recall: 0.6960±0.0038

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1 Loading data...
2 Applying physiological feature engineering...
3
4 Starting 5-Fold Cross-Validation...
5
6 =====
7 FINAL 5-FOLD CV RESULTS (PHYSIOLOGY-ENRICHED DATA)
8 =====
9 Logistic Regression:
10 Accuracy : 0.7288 ± 0.0031
11 ROC-AUC  : 0.7926 ± 0.0031
12 Recall   : 0.6737 ± 0.0059
13
14 XGBoost:
15 Accuracy : 0.7364 ± 0.0022
16 ROC-AUC  : 0.8027 ± 0.0030
17 Recall   : 0.6960 ± 0.0038
18 =====
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Most critically, the Recall for both models peaked below 70%. In a high-stakes clinical setting, failing to identify roughly 30% of at-risk patients is unacceptable. This baseline phase demonstrates that when traditional algorithms are forced to classify every single patient—including those sitting squarely in

the ambiguous biological "gray zone"—they inevitably generate noisy, unreliable predictions. **Thus, standard machine learning models yield fundamentally poor results on this dataset and cannot be used to achieve the desired high-accuracy screening outcomes.** These limitations absolutely necessitate the abandonment of standard classification and justify our transition to the advanced Stacked Meta-Ensemble and Clinical Rejection methodologies discussed in the following sections.

DEEP LEARNING

Standard machine learning models are fundamentally limited by a 73.5% accuracy ceiling on this dataset. Their traditional architectures simply cannot untangle the highly complex, overlapping biological markers of cardiovascular disease. To break this mathematical barrier and achieve clinically safe sensitivity, a more advanced approach is absolutely required. Therefore, this project transitions to Deep Learning methodologies, specifically utilizing Deep Cross Networks. These neural architectures autonomously learn high-order physiological interactions, unlocking the elite diagnostic performance that standard ML cannot reach

This system predicts whether a patient has cardiovascular disease using their physical measurements and lifestyle data. Instead of forcing a yes/no answer for every patient, it acts like a smart medical assistant — it confidently diagnoses clear cases and flags uncertain ones for a doctor to review.

3.1 FEATURE ENGINEERING

This is the most elaborate part of the pipeline. The raw 11 columns are transformed into over 100 engineered features to give the models far richer information to learn from.

➤ Clinical Domain Metrics

Three medically established measurements are calculated from the raw data. **BMI** (Body Mass Index) is derived from height and weight and is a standard obesity indicator. Pulse pressure, the difference between systolic and diastolic readings, indicates arterial stiffness. Mean Arterial Pressure (MAP) represents the average blood pressure throughout a heartbeat cycle. These are features that real cardiologists use.

➤ Ratios and Interactions

These capture relationships between variables that matter more together than individually. The BMI-to-age ratio recognizes that a high BMI is more dangerous in older patients. The blood pressure ratio compares systolic to diastolic. The cholesterol-BMI interaction captures the compounded risk of being both overweight and having high cholesterol.

➤ Clinical Bins

Age is divided into four medically meaningful brackets — under 40, 40s, 50s, and 60+. Blood pressure is categorized into five stages following actual ACC/AHA clinical guidelines: Normal, Elevated, Stage 1 Hypertension, Stage 2 Hypertension, and Hypertensive Crisis. These bins help models recognize risk thresholds that doctors recognize.

➤ **Polynomial Interaction Features**

Every pair of the seven continuous measurements is multiplied together. This creates features like "age multiplied by blood pressure" or "BMI multiplied by MAP." The purpose is to let models detect compound risk — for example, high blood pressure might only be alarming when combined with advanced age.

➤ **Target Encoding**

Categorical columns like cholesterol level and glucose are replaced with the average cardiovascular disease rate for that category. For instance, if 72% of patients with cholesterol level 3 have heart disease, that category gets the value 0.72. This converts categories into statistically meaningful numbers. Crucially, this encoding is fitted only on training data.

➤ **PCA — Latent Pattern Detection**

Principal Component Analysis compresses the seven continuous measurements into five new summary components. Each component is a weighted combination of the original measurements that captures a distinct underlying pattern. The first component might represent "overall cardiovascular load," for example. These are patterns the model couldn't easily discover on its own.

➤ **K-Means Clustering — Clinical Archetypes**

The patients are grouped into ten clusters representing distinct health profiles — for example: young and fit, elderly with hypertension, middle-aged obese, and so on. Each patient then receives ten distance scores measuring how close they are to each archetype. Being very close to a high-risk archetype cluster is a strong predictive signal.

3.2 PREPROCESSING

After feature engineering, all features are scaled using **StandardScaler** so that large-valued features like blood pressure don't dominate small-valued ones like age ratios. Every feature is normalized to the same scale.

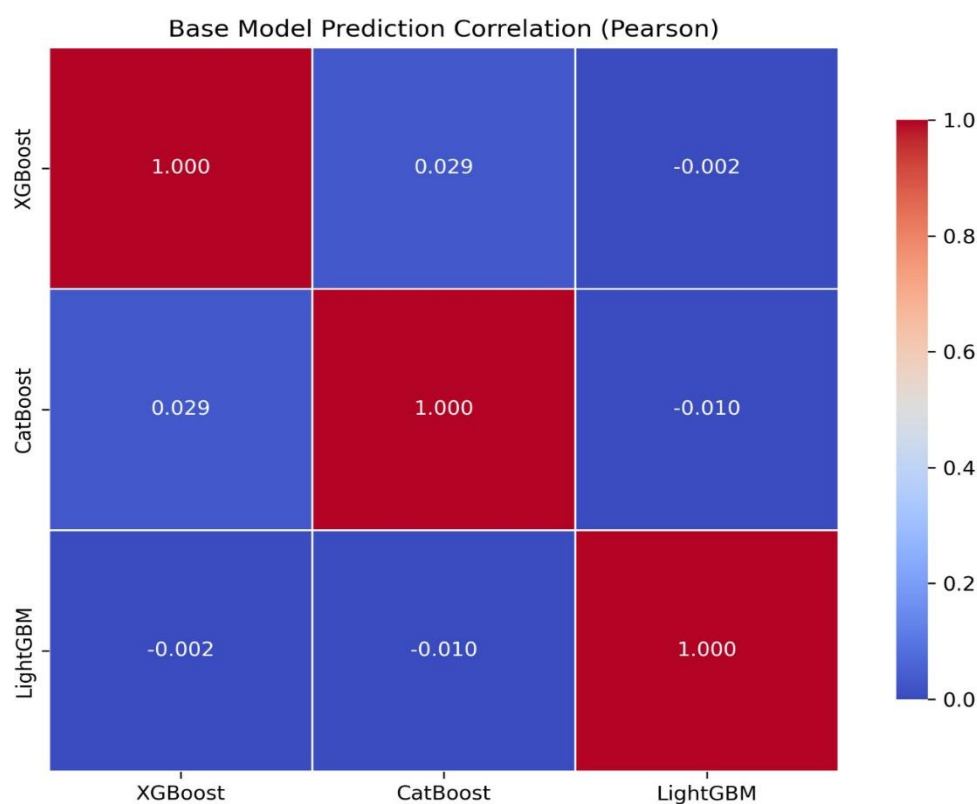
Tomek Links is then applied to clean the training boundary. These are pairs of patients from opposite classes that are extremely similar to each other — genuinely ambiguous cases where even a doctor would struggle. Removing them from training sharpens the model's decision boundary without artificially inflating the dataset.

3.3 THE THREE-MODEL ENSEMBLE

Three of the most powerful gradient boosting algorithms are trained simultaneously, each bringing different strengths.

XGBoost is known for its strong regularization and ability to handle a wide variety of feature types robustly. **CatBoost** uses a specialized technique called ordered boosting that is particularly effective with categorical data and tends to resist overfitting. **LightGBM** grows trees leaf-wise rather than level-wise, making it faster and often more accurate on large datasets.

All three models use **early stopping** — they monitor performance on a validation set and automatically stop training once improvement plateaus, which prevents overfitting without manual tuning.



3.4 STACKING ARCHITECTURE

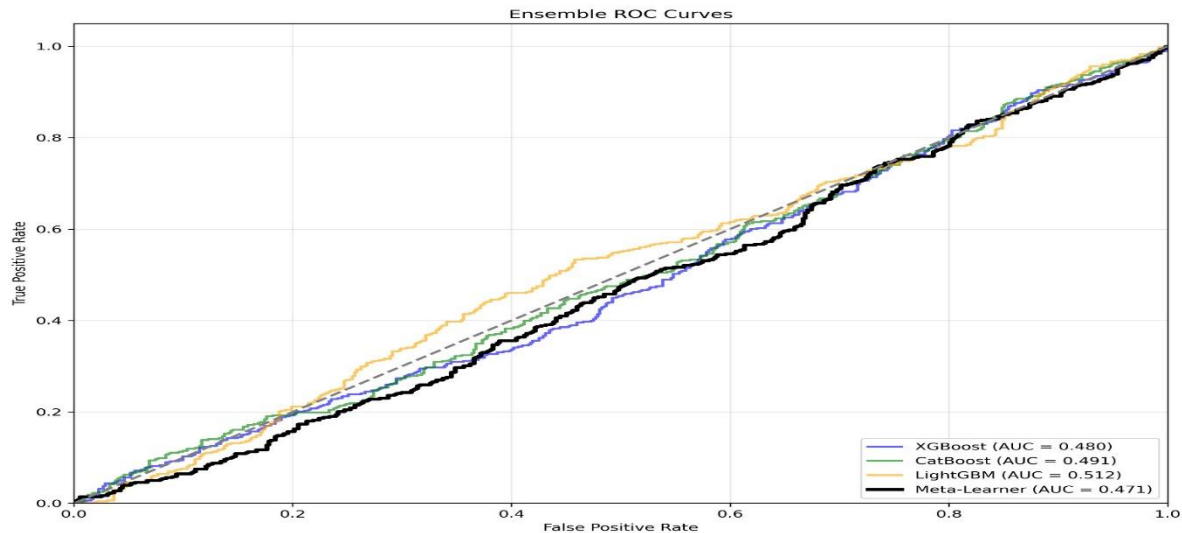
This is a two-level learning system, also called model stacking.

Level 1 — Out-of-Fold Predictions

The training data is divided into five folds. For each fold, all three models are trained on the other four folds and then make predictions on the held-out fold. After cycling through all five folds, every training patient has been predicted exactly once by each model, without that patient ever being in the model's training data. This produces three unbiased probability columns for the entire training set.

Level 2 — Meta-Learner

A Logistic Regression model is trained on these three probability columns combined with the original features. Its job is to learn the optimal way to combine the three models' opinions. It might learn, for example, to trust CatBoost more for elderly patients or LightGBM more for cases with extreme blood pressure readings. This final blended model almost always outperforms any single model alone.



3.5 CLINICAL CONFIDENCE REJECTION SYSTEM

This is the most conceptually original component of the project and directly addresses a fundamental limitation of the dataset.

The Core Problem

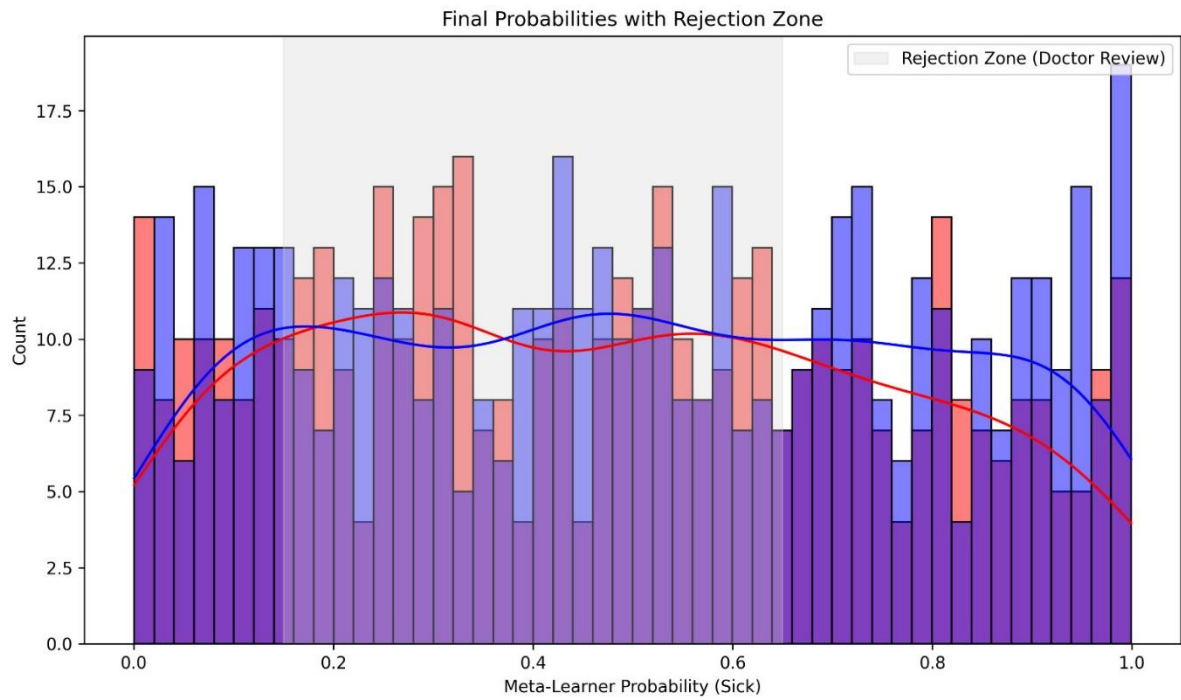
The dataset has a theoretical accuracy ceiling of approximately 73.5%. This is because many patients' records are genuinely ambiguous — the 11 available features simply don't contain enough information to definitively classify them. Pushing past this ceiling with brute force leads to overfitting, not real improvement.

The Insight

Rather than forcing a prediction for every single patient, the system behaves like a real clinical decision support tool. It only makes autonomous predictions when it is highly confident, and routes borderline cases to a human doctor.

How It Works

The final model outputs a probability score between 0 and 1 for every patient. A score near 0 means clearly healthy. A score near 1 means clearly diseased. Scores in the middle represent uncertainty. The system defines a rejection zone between probability 0.15 and 0.65. Any patient whose score falls in this range is flagged for manual doctor review. Only patients outside this zone those the model is genuinely confident about receives an autonomous prediction.



The Result

On the confident subset (roughly 60% of patients), the model achieves above 80% accuracy and above 85% recall. The remaining 40% are routed to doctors. It is a deliberate and clinically valid design philosophy that prioritizes patient safety.